Hi team, here is a quick overview of the **SunTax** platform for your testing and review. This guide covers where the application is hosted, the test accounts you can use, a simple 5-step walkthrough to test the full declaration flow, and our recommendations for production deployment and storage in Switzerland.

1. Live Deployment & Repository Links

Component	URL / Location	Notes
Web App (Frontend)	**https://frontend-ruby-gamma-46.vercel.app/**	Hosted on Vercel. Fully responsive across desktop and mobile.
Backend API	**https://suntax-api.onrender.com**	FastAPI service handling calculations, OCR, and user sessions.
API Docs (Swagger)	**https://suntax-api.onrender.com/api/docs**	Interactive API explorer to inspect all endpoints and test requests.
GitHub Repository	**https://github.com/usamashaikh13/suntax**	Full codebase: Next.js frontend, Python backend, and Swiss tax rules.

Note on initial loading: The backend is currently running on Render's free tier. If the server has been idle, the very first request takes roughly 45–60 seconds to spin up. All subsequent clicks and API calls will respond immediately.

2. Demo Accounts & Logins

Account	Credentials	2FA Setup & Access
Primary Test Account Full 2FA enabled	**Email:** client@suntax.ch **Password:** TestPassword123!	Requires a 6-digit 2FA code. • **Authenticator Key:** OWZDANSQXHFBZXANGHT3ZKUYM2XW2AAV • Or run: python get_2fa.py client@suntax.ch
Standard Demo Account Direct sign-in	**Email:** demo@suntax.ch **Password:** TestPassword123!	Logs in directly without 2FA prompt. You can test setting up 2FA on your phone from *My Account → Security*.

3. How to Test the Application (5-Step Walkthrough)

1. Sign In	Go to the live web app and log in with **client@suntax.ch** / **TestPassword123!**. Enter the 6-digit code from your authenticator app when prompted.
2. Start Return	On your dashboard, click **+ New Tax Return**. Select Canton **ZH (Zürich)**, Municipality **Zürich (BFS #261)**, and Tax Year **2025**, then click confirm.
3. Upload Documents	In the **Documents** tab, drag and drop test documents (e.g. Salary Certificate, Bank Statement, Pillar 3a Certificate, or Donation Receipt). The OCR worker extracts financial data automatically.
4. Review Extracted Data	Click **Review Fields** on any processed document to verify the extracted figures, then click **Approve & Apply to Profile** to save them to your tax declaration.
5. Calculate & Submit	Go to the **Calculation** tab to run the official Swiss tax engine (Federal, Cantonal, and Municipal breakdown). When ready, proceed to **Review & Submit** and finalize the declaration.

4. Database & Codebase Details

• **Local SQLite Database:** Located at backend/suntax_dev.db. It comes pre-seeded with tax rules for initial cantons, test user accounts, and sample declarations so the app works out-of-the-box locally.

• **Production PostgreSQL 16:** In production, the system connects via asyncpg and uses Row-Level Security (RLS) policies. Each user's documents and declarations are isolated at the database level.

• **Document File Storage:** Uploaded files are stored in backend/dev_storage/ during development (mirroring an S3 bucket structure with user-prefixed paths). In production, this maps to an S3-compatible bucket.

• **Running Locally:** Simply run ./backend/start_dev.sh to create the Python virtualenv and start the backend, and npm run dev inside frontend/ to run the web interface.

5. Core API Endpoints Reference

Endpoint	Method	Purpose
`/api/v1/auth/login`	POST	User authentication; returns JWT token or 2FA challenge
`/api/v1/auth/login/2fa`	POST	Validates 6-digit TOTP code and returns full session token
`/api/v1/tax-returns`	GET / POST	Lists all user tax returns or creates a new draft declaration
`/api/v1/documents/upload`	POST	Uploads tax documents, generates hash, and triggers OCR pipeline
`/api/v1/tax-returns/{id}/calculate`	POST	Runs deterministic Swiss tax calculation for federal, canton & city
`/api/v1/tax-returns/{id}/confirm`	POST	Finalizes the tax declaration with 2FA verification
`/api/v1/tax-returns/{id}/export/pdf`	POST	Generates official tax declaration summary PDF for download

6. Production Hosting & Storage Recommendations

Swiss tax declarations contain sensitive personal and financial information. When moving to production, infrastructure choices should comply with the **Swiss Federal Act on Data Protection (FADP / nDSG)** and ensure data stays within Switzerland.

A. Document Storage (Tax Certificates & PDFs)

• **Top Recommendation — Infomaniak Swiss S3 (Geneva / Winterthur):** 100% Swiss company and infrastructure, ISO 27001 certified, and immune to foreign laws like the US Cloud Act. Costs around CHF 0.012 / GB / month with no egress fees.

• **Alternative — Exoscale Object Storage (Zürich / Geneva):** Enterprise Swiss cloud provider built for financial institutions. Fully S3-compatible with high speed and zero bandwidth fees inside the Swiss zone.

• **Enterprise Standard — AWS S3 Zürich (eu-central-2):** A reliable option if your team already operates on AWS. Guarantees Swiss data residency and supports customer-managed encryption keys (KMS).

B. Application & Database Hosting

• **Frontend Web App — Vercel Pro (~$20 / mo):** Provides global edge delivery, automatic HTTPS, preview builds for PRs, and fast load times for Swiss users with zero server maintenance.

• **Backend API & OCR Worker — Exoscale Cloud VPS or SKS (Zürich, ~CHF 38 / mo):** Runs the FastAPI backend and document OCR processes directly on servers in Zürich, keeping all data processing in Switzerland.

• **Database — Exoscale Managed PostgreSQL 16 (Zürich, ~CHF 45 / mo):** Managed database with automatic daily backups, point-in-time recovery, and encryption at rest.

C. Recommended Production Stack & Estimated Monthly Cost

Component	Provider & Region	Est. Cost
Frontend Web App	Vercel Pro (Global Edge)	~$20 / mo
Backend API & Worker	Exoscale Cloud VPS (Zürich, Switzerland)	~CHF 38 / mo
Managed PostgreSQL	Exoscale Managed DB (Zürich, Switzerland)	~CHF 45 / mo
Secure Document Vault	Infomaniak Swiss S3 (Geneva / Winterthur)	~CHF 5–10 / mo
Total Monthly Cost	**100% Swiss Data Residency Stack**	**~CHF 110–120 / mo**